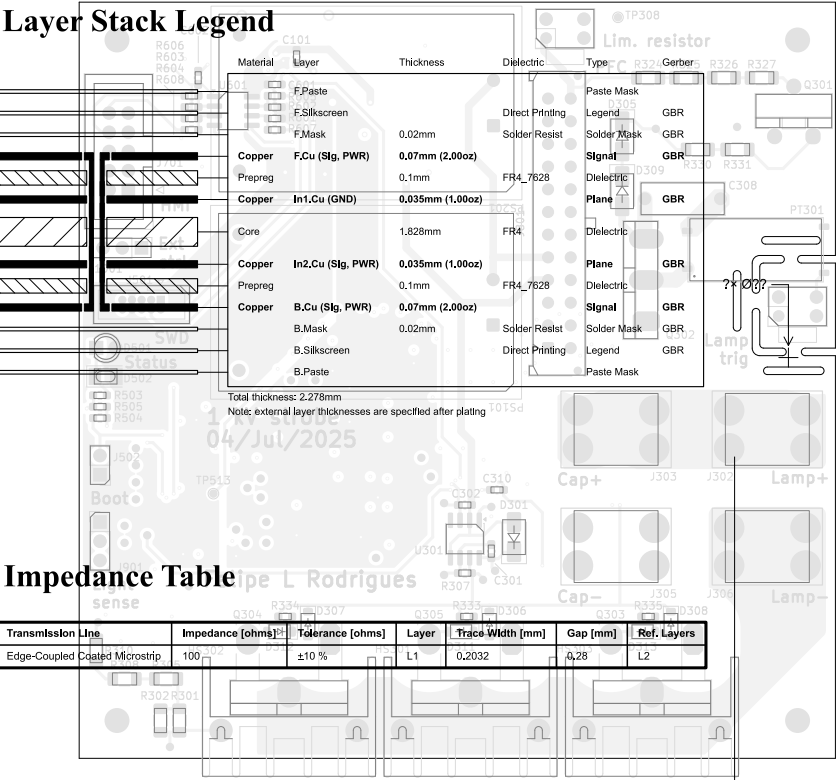


Main board Fabrication Document

Top Fabrication (Scale 1:1)

Layer Stack Legend



Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	100	±10 %	L1	0.2032	0.28	L2

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 4) SURFACE FINISH: HAL SNPB
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR GREEN.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:

A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
B. Tg 170 C OR EQUIVALENT.
C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY CHURRASCOD ELECTRONICS.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE 100,000 × 100,000 mm
BOARD THICKNESS 2.278 mm
TRACE WIDTH 0.300 mm
TRACE TO TRACE 0.200 mm
MIN. HOLE (PTH) 0.300 mm
MIN. HOLE (NPTH) 1.000 mm
ANNULAR RING 0.100 mm
COPPER TO HOLE 0.300 mm
COPPER TO EDGE 0.500 mm
HOLE TO HOLE 0.250 mm

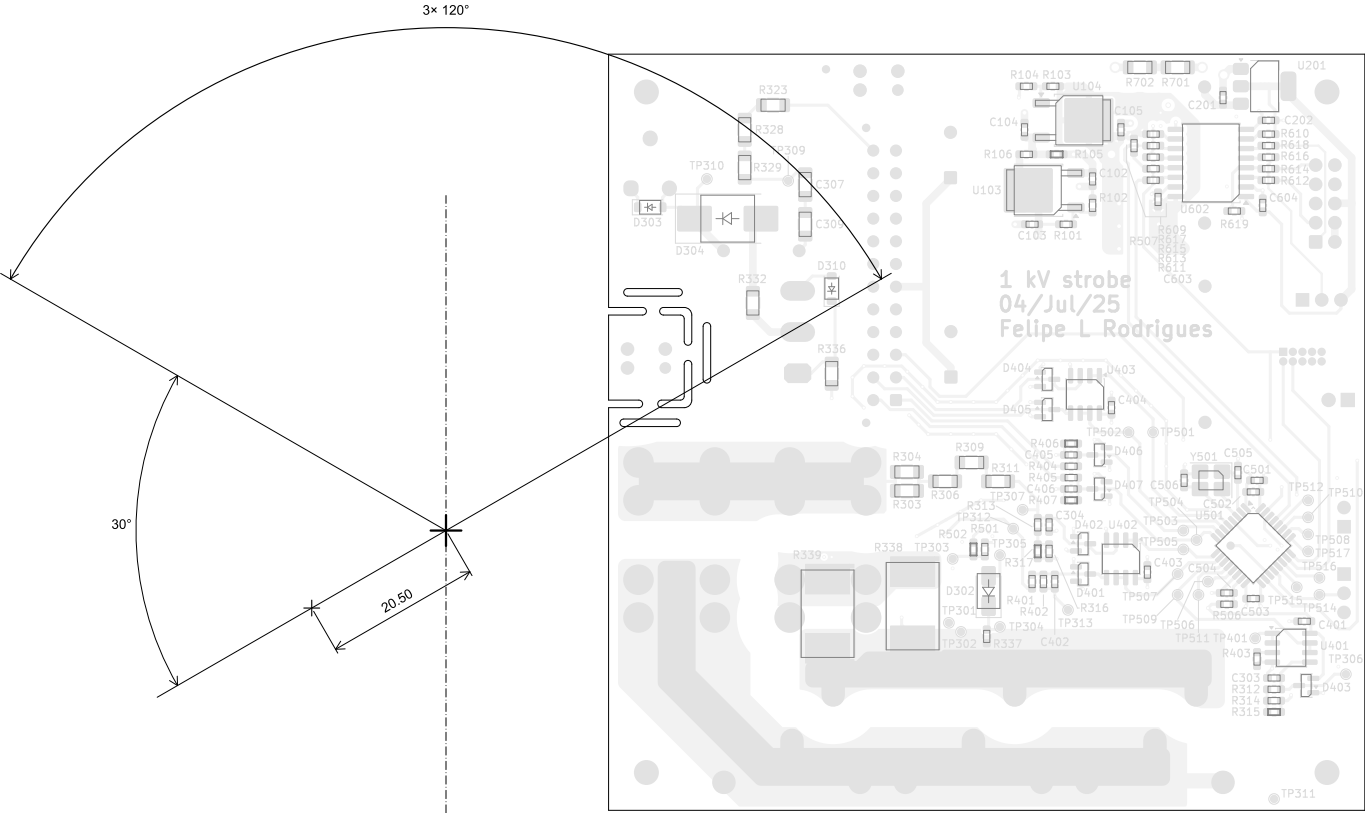
All dimensions are in millimeters unless otherwise specified.



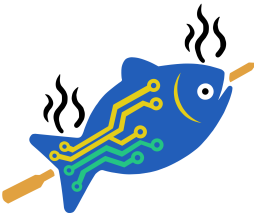
Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 272ff09
	Board Name: Main board		Project Name: 1 kV strobe	
Sheet Title: Top Fabrication (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 1 of 7

Main board Fabrication Document

Bottom Fabrication (Scale 1:1)



All dimensions are in millimeters unless otherwise specified.



ChurrasCod
Electronics

Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY		Git Hash: 272ff09	
	Board Name: Main board		Project Name: 1 kV strobe			
Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)		
Sheet Path:		Reviewer:	Size: A4	Sheet: 2 of 7		

Main board Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	109	0,30mm (11,81mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Via
○	12	0,50mm (19,69mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Via
+	31	0,50mm (19,69mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
□	10	0,65mm (25,59mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
◇	2	0,80mm (31,50mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊠	2	0,98mm (35,43mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
※	4	1,00mm (39,37mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Via
⊗	3	1,00mm (39,37mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊙	24	1,02mm (40,16mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊕	3	1,10mm (43,31mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊖	6	1,20mm (47,24mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊗	9	1,56mm (59,06mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊕	16	1,80mm (70,87mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
	Total 262					



Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY		Git Hash: 272ff09	
	Board Name: Main board		Project Name: 1 kV strobe			
Sheet Title: Drill Drawing (L1 - L4)	File Name: Strobe_main.kicad_pcb		Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)	
Sheet Path:			Reviewer:	Size: A4	Sheet: 3 of 7	

Main board Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

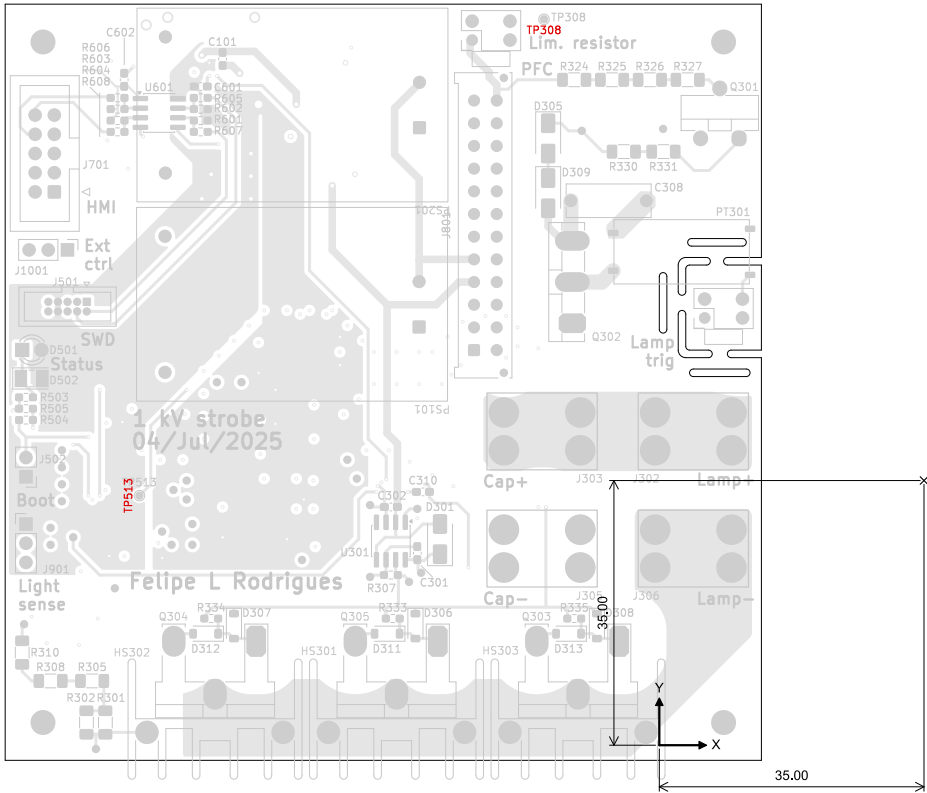
Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	2	1,02mm (40,16mils)	NPTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Mechanical
○	2	1,35mm (53,15mils)	NPTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Mechanical
+	4	3,20mm (125,98mils)	NPTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Mechanical
Total 8						



Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 272ff09
	Board Name: Main board		Project Name: 1 kV strobe	
Sheet Title: Drill Drawing (L1 - L4)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 4 of 7

Main board Fabrication Document

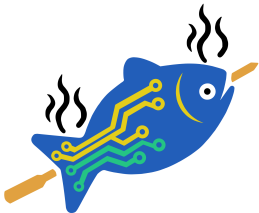
Top Test Points (Scale 1:1)



Ref.	Net	X [mm]	Y [mm]
TP308	LAMP_P	34.25	77.00
TP513	Vbus_ctrl	-19.25	14.00

Ref.	Net	X [mm]	Y [mm]
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All dimensions are in millimeters unless otherwise specified.



ChurrasCod
Electronics

Comments:

Company:

ChurrasCod Electronics

Variant:

PRELIMINARY

Git Hash:

272ff09

Board Name:

Main board

Project Name:

1 kV strobe

Sheet Title:

Top Test Points (Scale 1:1)

File Name:

Strobe_main.kicad_pcb

Designer:

FR

Date:

2024-04-13

Revision:

+ (Unreleased)

Sheet Path:

Reviewer:

Size:

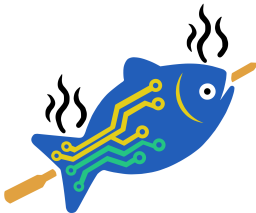
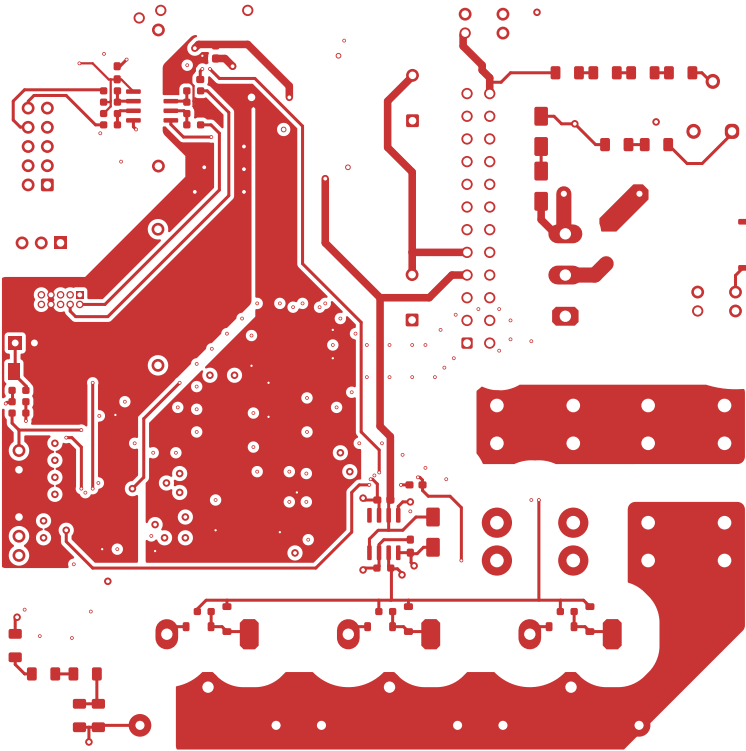
A4

Sheet:

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Main board Fabrication Document

F.Cu (Scale 1:1)

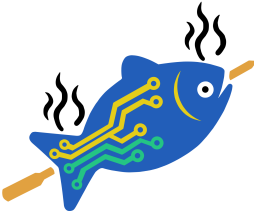
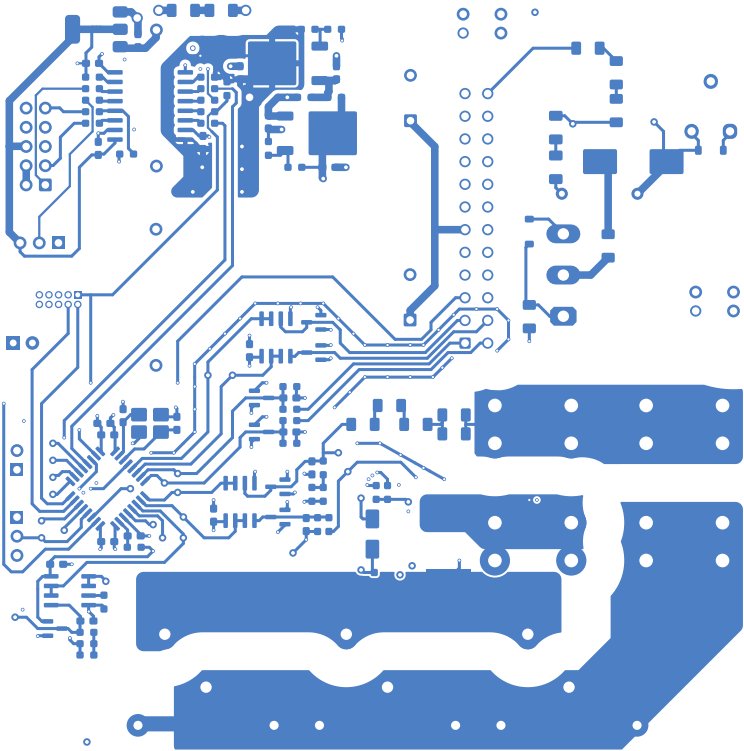


ChurrasCod
Electronics

Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 272ff09
	Board Name: Main board		Project Name: 1 kV strobe	
Sheet Title: F.Cu (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 6 of 7

Main board Fabrication Document

B.Cu (Scale 1:1)



ChurrasCod
Electronics

Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 272ff09
	Board Name: Main board		Project Name: 1 kV strobe	
	Sheet Title: B.Cu (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13
Sheet Path:			Revision: + (Unreleased)	
			Size: A4	Sheet: 7 of 7